

Azure Service Bus – Abandon a Message

1. What is Abandon?

When a receiver **abandons** a message:

- The message **remains in the queue**.
- The message becomes available for processing again.
- The **delivery count increases by 1**.

Easy Definition

Abandon = Do not process this message now, make it available again.

2. Practical Example

In Azure Service Bus Explorer, the queue contains a message:

Message to be ignored by receiver

Initially, its delivery count is:

Delivery Count = 0

The application receives this message but decides not to process it.

Instead of completing it, the application abandons it.

3. Code for Abandoning a Message

After receiving the message:

```
ServiceBusReceivedMessage message =  
    await receiver.ReceiveMessageAsync();
```

Abandon it using:

```
await receiver.AbandonMessageAsync(message);
```

The important line is:

```
await receiver.AbandonMessageAsync(message);
```

This tells Azure Service Bus:

"I am not completing this message. Make it available again."

4. What Happens to Delivery Count?

Before abandoning:

```
Message:  
"Message to be ignored by receiver"  
  
Delivery Count = 0
```

After abandoning:

```
Message:  
"Message to be ignored by receiver"  
  
Delivery Count = 1
```

The message is **still present in the queue**.

So:

```
Before:  
Queue → Message → Delivery Count: 0  
  
After Abandon:  
Queue → Message → Delivery Count: 1
```

5. Why Does Delivery Count Increase?

Azure Service Bus has already attempted to deliver the message to a receiver.

The receiver received the message but abandoned it.

Therefore, the next delivery is considered another delivery attempt.

```
First delivery
  ↓
Receiver receives message
  ↓
Receiver abandons message
  ↓
Message becomes available again
  ↓
Delivery Count increases
```

Important

Abandon does not delete the message.

It only makes the message available again.

6. Practical Verification

We can verify this behavior using Azure Portal.

Before Abandon

Open:

Queue → Service Bus Explorer → Pick from start

You will see:

```
Message: Message to be ignored by receiver
Delivery Count: 0
```

Run the C# application:

```
await receiver.AbandonMessageAsync(message);
```

Then refresh the queue.

After Abandon

The same message is still present:

```
Message: Message to be ignored by receiver  
Delivery Count: 1
```

This confirms that:

- The message was **not deleted**.
- The message was **abandoned**.
- The delivery count increased from **0 → 1**.

7. Complete vs Abandon

Operation	Message	Delivery Count
Complete	Removed from queue	No further delivery
Abandon	Remains available	Increases

Easy Memory Trick

Complete → Delete

Abandon → Keep + Retry

8. When Would We Use Abandon?

Use **Abandon** when the current receiver cannot or should not process the message and you want the message to become available for another processing attempt.

For example:

```
Receive Message  
↓
```

```
Cannot process
  ↓
Abandon
  ↓
Message available again
  ↓
Another delivery attempt
```

9. Important Interview Question

Q: What happens when you abandon an Azure Service Bus message?

Answer:

When a message is abandoned, it remains in the queue and becomes available for another delivery attempt. Its delivery count is incremented because Service Bus has attempted to deliver the message to a receiver.

10. Quick Revision

```
Receive Message
  ↓
Do not complete it
  ↓
AbandonMessageAsync()
  ↓
Message remains in Queue
  ↓
Delivery Count + 1
  ↓
Available for another delivery
```

Code to Remember

```
await receiver.AbandonMessageAsync(message);
```

Abandon = Message stays in the queue + delivery count increases + message can be delivered again.